Exam Focus Plus

FY-926

Reg. No. : .....

Name : .....

FIRST YEAR HIGHER SECONDARY
IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

BIOLOGY

(Botany & Zoology)

Maximum : 60 Scores

Time : 2 Hours

Cool-off time : 15 Minutes

Preparatory Time : 10 Minutes

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time. Further there is a '10 minutes' 'Preparatory Time' at the end of the Botany Examination and before the commencement of Zoology Examination.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും. കൂടാതെ ബോട്ടണി പരീക്ഷയ്ക്കുശേഷം സുവോളജി പരീക്ഷ തുടങ്ങുന്നതിനുമുമ്പ് '10 മിനിറ്റ്' തയ്യാറെടുപ്പുകൾ നടത്തുന്നതിനായി നല്കുന്നതാണ്.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

Exam Focus Plus

PART – A

BOTANY

(Maximum : 30 Scores) **Time : 1 Hour**

I. Answer any 3 questions from 1 to 5. Each question carries 1 score. **(3 × 1 = 3)**

1. Nitrogen fixing cells seen in Anabaena is called _____

2. Which of these algae produce the hydrocolloidal substance called carrageen ?

   (a) Green algae (b) Brown algae

   (c) Red algae (d) All of these

3. If a fruit is formed without fertilisation of the ovary, it is called a _____ fruit.

4. The process of division of nucleus during cell division is known as _____ .

5. Which is the first isolated naturally existing cytokinin ?

II. Answer any 9 questions from 6 to 16. Each question carries 2 scores. **(9 × 2 = 18)**

6. (a) Which are the two substances formed as result of breakdown of fat during respiration ?

   (b) It would be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one. Why ?

7. The life cycle of Bryophytes has two stages.

(a) Which are they ?

(b) Write one difference between them.

8. By observing the functions of plant hormones given in the first column, write the name of plant hormone that is responsible for this function in second column.

Function of plant hormone	Name of plant hormone
(a) Ripening of fruits	(i) .....
(b) Bolting	(ii) .....
(c) Inducing seed dormancy	(iii) .....
(d) Apical dominance	(iv) .....

9. (a) Name the class of fungi commonly known as sac fungi.

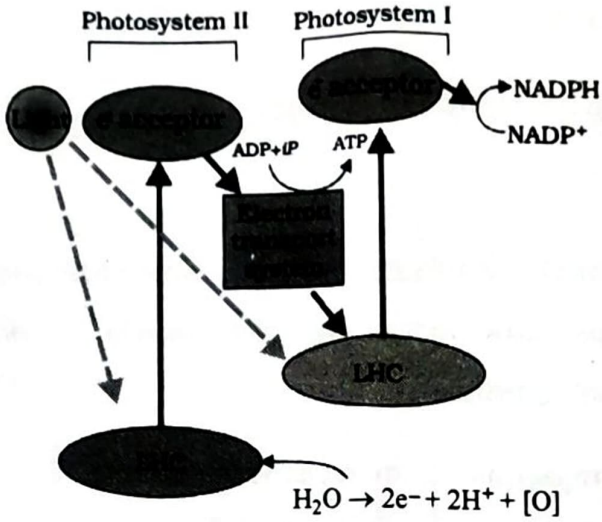
(b) Which fungus of this class is extensively used for biochemical and genetical studies ?

10. Write one difference between the following :

(a) Open and Closed vascular bundle

(b) Conjoint and Radial vascular bundle

11. Observe the given figure and answer the following questions :



(a) Name the process.

(b) Where does it take place in Chloroplast ?

(c) What is the significance of splitting of water in this process ?

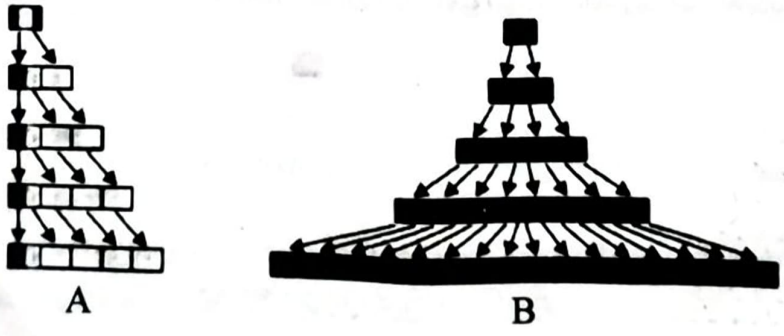
12. (a) Write the name of any two cell organelles bounded by two membranes.

(b) Write one function of each.

13. Observe the figures A and B and answer the following questions :

(a) Which are the two types of growth rates represented in figure A and B ?

(b) Write the difference between these two growth rates.



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14. In grasses, how Bulliform cells helps to minimise water loss from leaves ?

15. (a) Why C4 plants have better productivity and yield than C3 plants ?

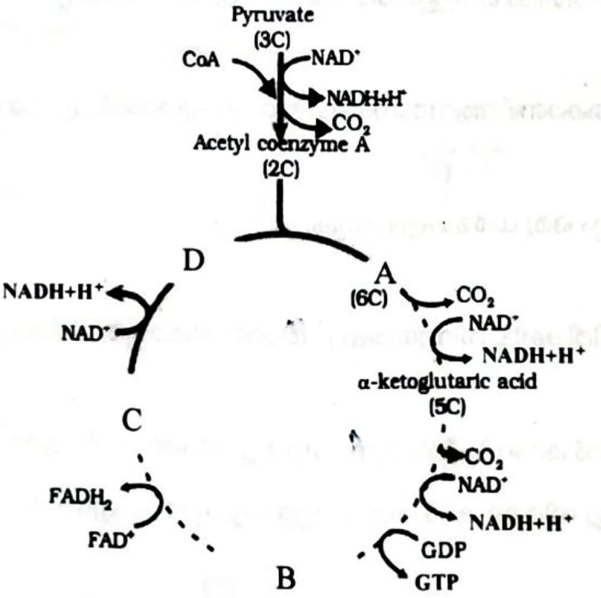
(b) Write two examples of C4 plants.

16. (a) The cell membrane is mainly consists of two substances. Which are they ?

(b) Write the difference between Passive transport and Active transport.

III. Write any 3 questions from 17 to 20. Each question carries 3 scores. **(3 × 3 = 9)**

17. A process of aerobic respiration is given in the picture. Observe the picture and answer the following questions :



(a) Name the process.

(b) Write the names of compounds marked as A,B,C and D in the picture.

(c) Where does this process occur in the cell ?

18. Briefly describe the following terms associated with monocotyledonous seed :

(a) Aleurone layer

(b) Scutellum

(c) Coleorhiza

19. Write three differences between C3 plants and C4 plants.

20. Some of the events of the stages of Mitosis is given below. Write them in their appropriate columns of the table :

(i) Spindle fibres attach to kinetochores of chromosomes

(ii) Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei

(iii) Centromeres split and chromatids separate

(iv) Chromosomal material condenses to form compact mitotic chromosomes

(v) Chromatids move to opposite poles

(vi) Centrosome which had undergone duplication during interphase, begins to move towards opposite poles of the cell

Pro phase	Meta phase	Ana phase	Telo phase

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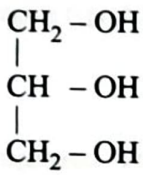
PART – B

ZOOLOGY

(Maximum : 30 Scores) **Time : 1 Hour**

I. Answer any 3 questions from 1 to 5. Each carries 1 score. **(3 × 1 = 3)**

1. Identify the biomolecule given below :



2. Expand ICZN.

3. Note the relationship between the first two words and find a suitable word for the fourth place.

RBC : Haemoglobin :: Muscles : _____

4. Name the centre present in the medulla region of brain that is primarily responsible for the regulation of respiration.

5. Name the hormone secreted by the heart wall which decreases blood pressure.

II. Answer any 9 questions from 6 to 16. Each carries 2 scores. **(9 × 2 = 18)**

6. Match the following :

A	B
Platyhelminthes	Pinnae
Birds	Radula
Mollusca	Flame cells
Mammals	Parapodia
	Pneumatic bones

7. Categorise the following terms into two columns as primary and secondary metabolites :

Amino acids, Flavonoids, Rubber, Sugars

8. (a) Name the three main plasma proteins found in human blood.

(b) Plasma without clotting factor is called

9. (a) Name the common chamber into which the digestive, reproductive, and excretory systems open in a frog.

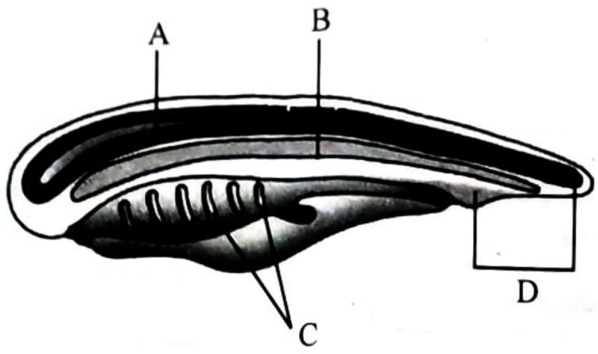
(b) Write any one distinguishing feature between a male and a female frog.

10. A specialised cardiac musculature called nodal tissue is distributed in the heart.

Prepare a flow chart showing the pathway of cardiac impulses through the following nodal tissues :

AVN, Purkinje fibres, AV bundle, SAN

11. Label the chordate characters A, B, C and D shown in the diagram :



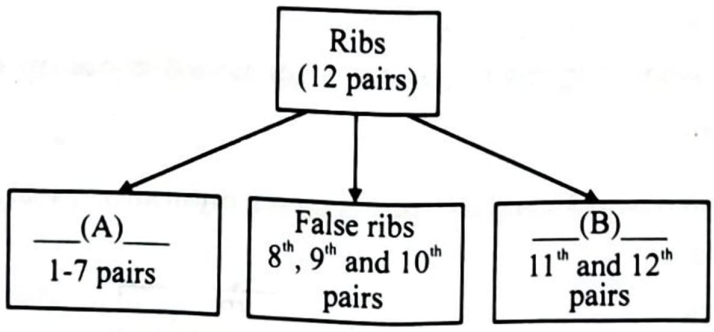
12. The non-protein part of an enzyme is called a cofactor. There are three types of cofactors : prosthetic groups, coenzymes and metal ions.

(a) Which type of cofactor is tightly bound to the enzyme ?

(b) The protein part of an enzyme is called apo.

(c) What happens when cofactor is removed from the enzyme ?

13. (a) Complete the chart showing the different types of ribs :



(b) Inflammation of joints due to the accumulation of uric acid crystals is called _____.

14. In ___(A)___ type of synapse, the membranes of the presynaptic and postsynaptic neurons are in close proximity.

In ___(B)___ type of synapse, the membranes of the presynaptic and postsynaptic neurons are separated by a fluid-filled space.

(a) Identify the types of synapses (A) and (B).

(b) Name the fluid-filled space present in synapse (B).

15.



(a) Identify the animal.

(b) Name the class to which it belongs.

16. Which hormonal deficiency is responsible for the following diseases ?

(a) Diabetes mellitus

(b) Goitre

(c) Diabetes insipidus

(d) Dwarfism

III. Answer any 3 questions from 17 to 20. Each carries 3 scores. **(3 × 3 = 9)**

17. Observe the picture and answer the following questions :

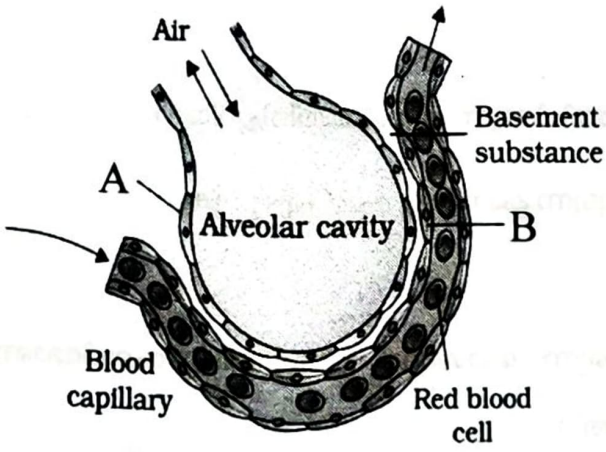


(a) Identify the picture.

(b) How does this structure help the organism ?

(c) Name the phylum to which it belongs.

18. Observe the diagram showing a section of an alveolus with a pulmonary capillary and answer the following questions :



(a) Label the diffusion membranes A and B.

(b) Write any two factors that affect the rate of diffusion across the diffusion membranes.

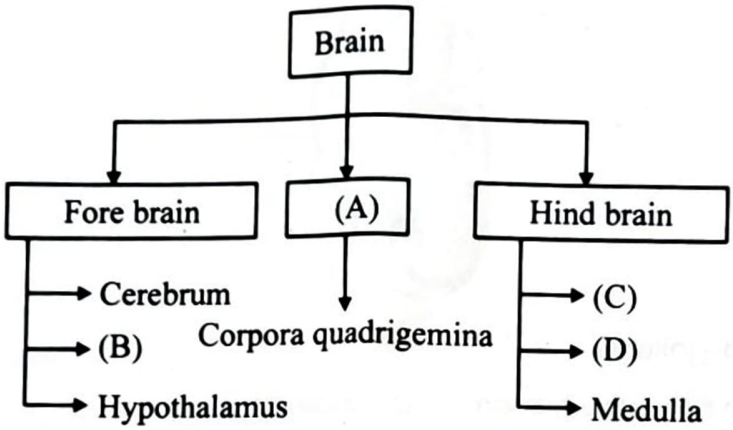
(c) What is the maximum number of oxygen molecules that each haemoglobin molecule can carry ?

19. (a) Which are the three main processes involved in urine formation ?

(b) Differentiate between cortical nephrons and juxta medullary nephrons.

(c) Inflammation of glomeruli of the kidney is called_____.

20. (a) Complete the chart :



(b) The two cerebral hemispheres are connected by a tract of nerve fibres called _____.